

EXAM Linear Control Design 1

31 May 2021 Questions - Multiple Choice

1. Q1. What is the transfer function of this linear system in Fig.1?

(1 Point)

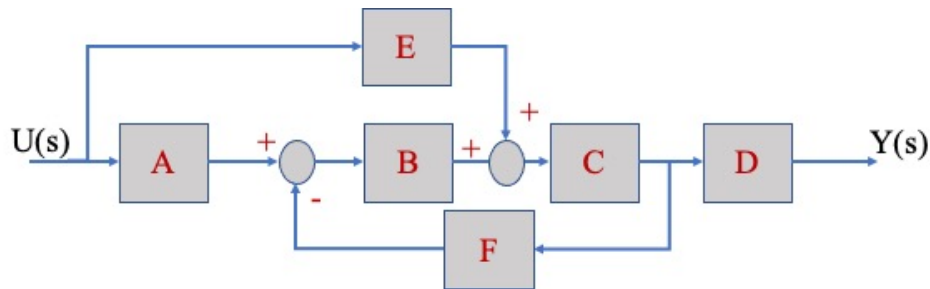


Figure 1: Block Diagram

1. $\frac{Y}{U} = \frac{ABD + DEF^2}{1 + BCF}$
2. $\frac{Y}{U} = \frac{ABCD + ECD}{1 + BCF}$
3. $\frac{Y}{U} = \frac{ABCD + EF}{BCF}$
4. $\frac{Y}{U} = \frac{ABCD + ECD + EC^2FBD}{1 + CDE}$
5. $\frac{Y}{U} = \frac{ABCD}{1 + BCF} + ECD$

Answer:

2. Q2. Consider an *RLC* circuit in Laplace domain as the one in Fig. 2.

What is the corresponding block diagram?

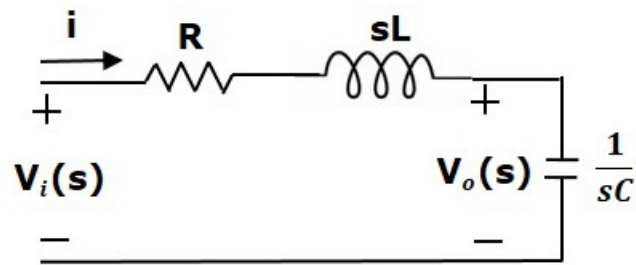


Figure 2: RLC circuit

- 1.
- 2.
- 3.
- 4.
- 5.

Answer:

3. Q3. A system has poles and zeros as shown in Fig. 3 (the circle marks the zero, the cross marks the pole). **Which is the corresponding Bode plot?** (1 Point)

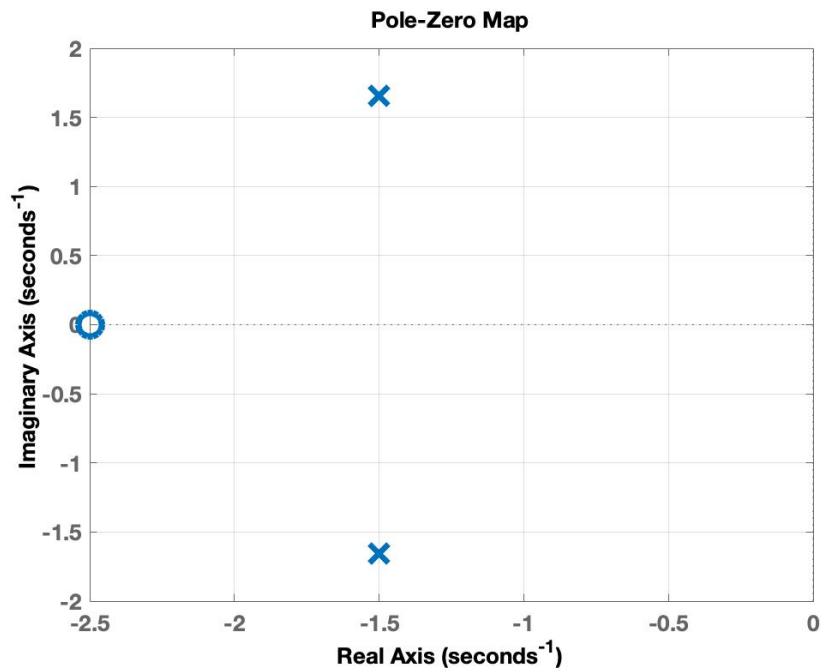
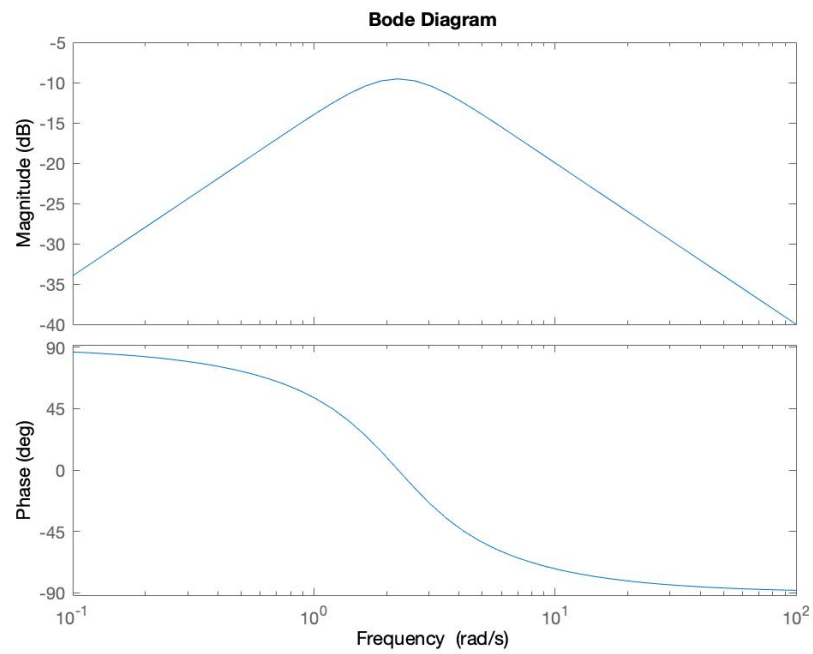
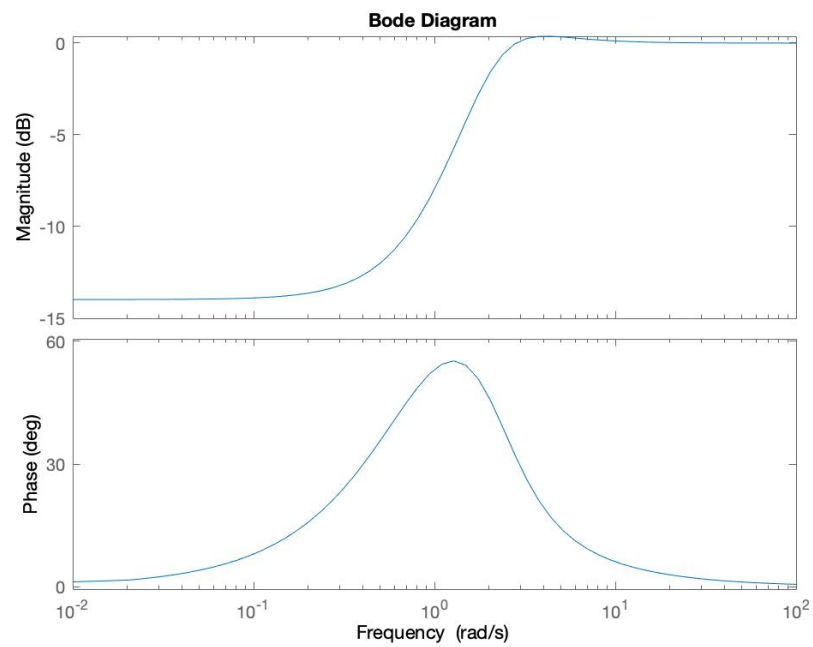


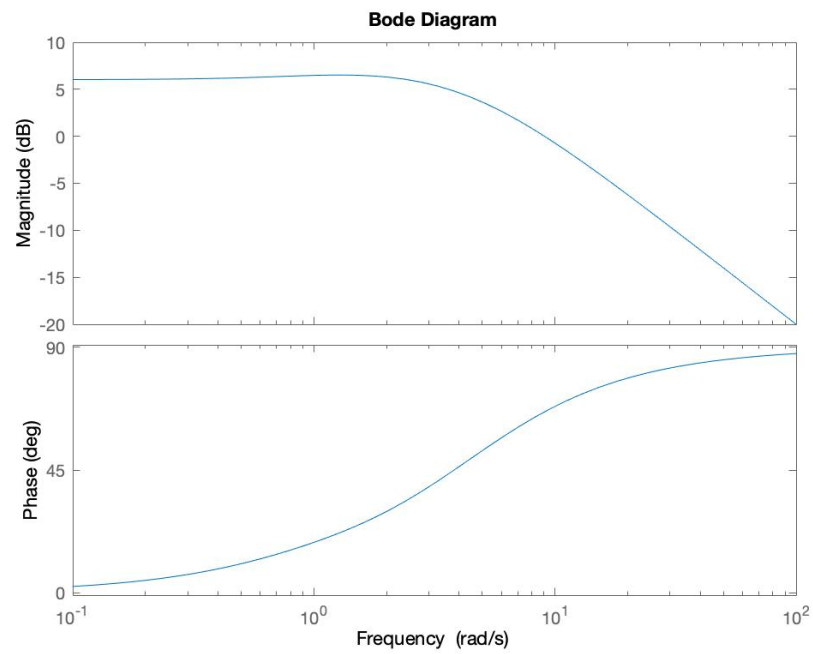
Figure 3: Poles/zeros



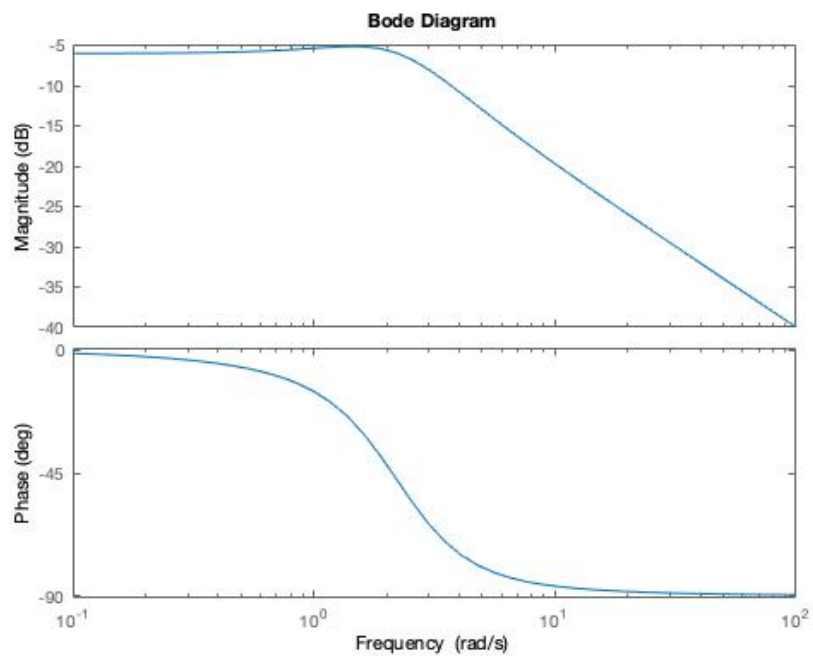
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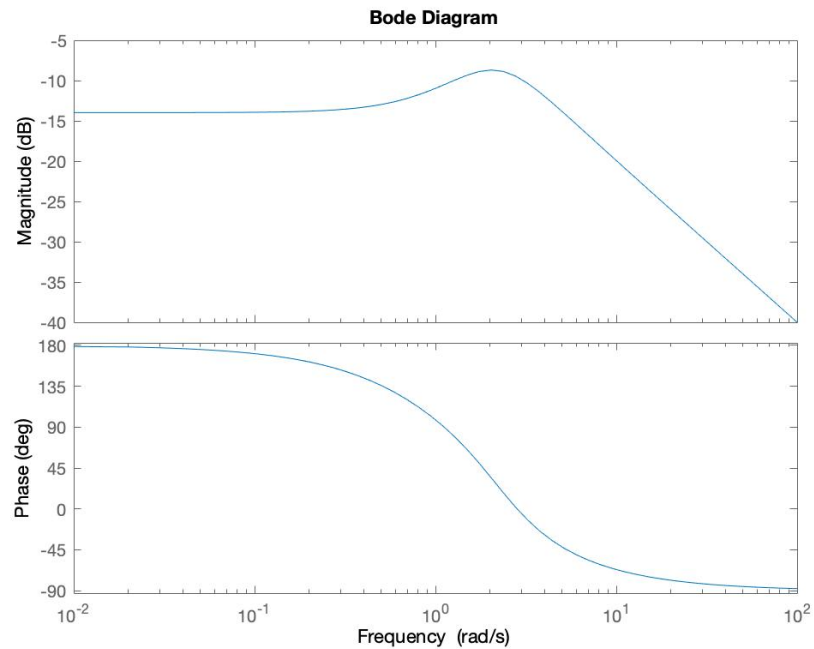
2.



3.



4.



5.

Answer:

4. Q4. The following closed-loop system is given in Fig. 4. Where

$$P(s) = \frac{1}{(s+1)^3}$$

is the system transfer function and K is a proportional controller with gain $K \in \mathbb{R}^+$.

For which values of K is the closed loop stable? (1 Point)



Figure 4: Closed-loop system

1. $K \geq 8$
2. $-1 < K < 6$
3. $0 < K < 8$

4. $K > -10$

Answer:

5. Q5. The bode plot diagrams of the magnitude and phase of a transfer function $G(s)$ of a linear system is shown in Fig. 5.

Which of the following answers is correct?

(1 Point)

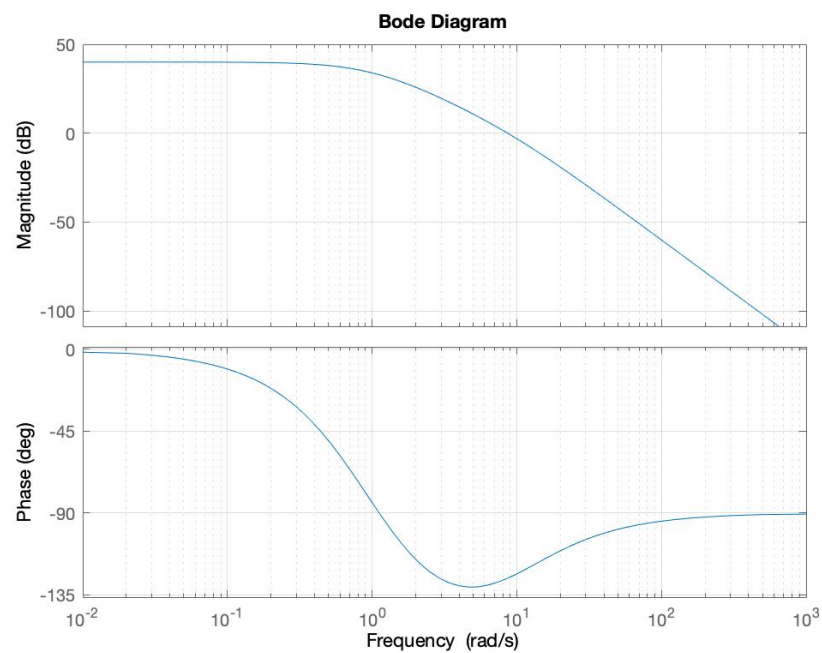


Figure 5: Bode Plot Diagrams - Q5

1. $G(s)$ has negative and positive real poles and no zeros
2. $G(s)$ has only 1 negative real pole and 1 negative real zero
3. $G(s)$ has only 1 positive real pole and no zeros
4. $G(s)$ has positive real poles and 1 negative real zero

Answer:

6. Q6. A closed-loop system has a loop transfer function

$$L(s) = \frac{K}{s(s+3)(s+10)}$$

and the following Bode plot in Fig.6

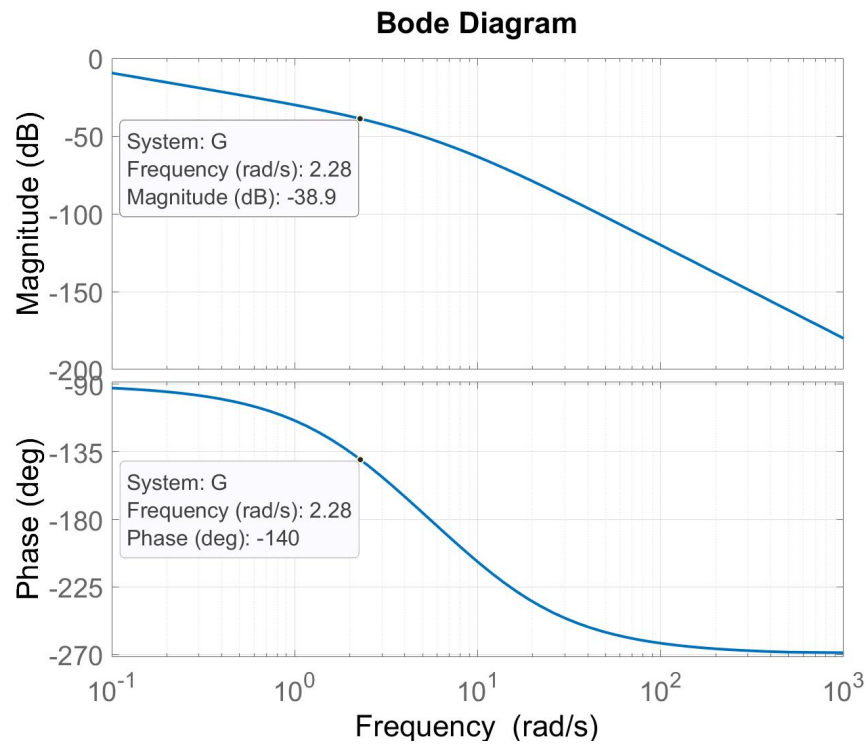


Figure 6: Bode Plot - Q6

What is the gain K so that the phase margin is $PM=40^\circ$?

(1 Point)

1. $K = 19.5$
2. $K = 90$
3. $K = 44$
4. $K = 38.9$
5. $K = 88 = 88.1$

Answer:

7. Q7. Which is the transfer function $\frac{X_2(s)}{F(s)}$ for the system shown in Figure 7? Both masses slide on a frictionless surface and $k = 1 \text{ N/m}$. (Consider zero initial conditions). [hint = write the equations of motion]

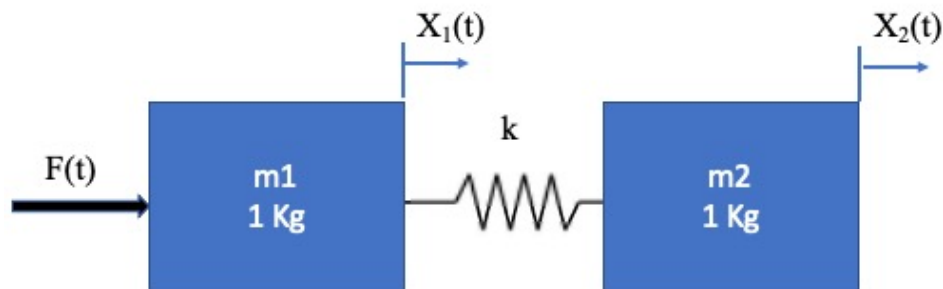


Figure 7: Two connected masses on a frictionless surface

(1 Point)

1. $\frac{X_2(s)}{F(s)} = \frac{1}{(s^2) * (s^2 + 1)}$
2. $\frac{X_2(s)}{F(s)} = \frac{1}{s^2 * (s^2 + 2)}$
3. $\frac{X_2(s)}{F(s)} = \frac{1}{s^2 * (s + 2)}$
4. $\frac{X_2(s)}{F(s)} = \frac{1}{s * (s^2 + 1)}$
5. $\frac{X_2(s)}{F(s)} = \frac{1}{(s^4 + 1)}$

Answer:

8. Q8. Consider the differential equation

$$\ddot{y}(t) + 2\dot{y}(t) + y(t) = u(t)$$

Where $y(0) = \dot{y}(0) = 0$ and $u(t)$ is the input.

Which are the poles of this system?

(1 Point)

1. $s_1 = 2, s_2 = -1$
2. $s_1 = -1, s_2 = -1$
3. $s_1 = 1j, s_2 = -1j$
4. $s_1 = -1, s_2 = -2$
5. $s_1 = -j, s_2 = -j$

Answer:

9. Q9. For the unit feedback system shown in Fig.8, **Which is the gain K of the proportional controller so that the output $y(t)$ has an overshoot of no more than 12% in response to a unit step?**

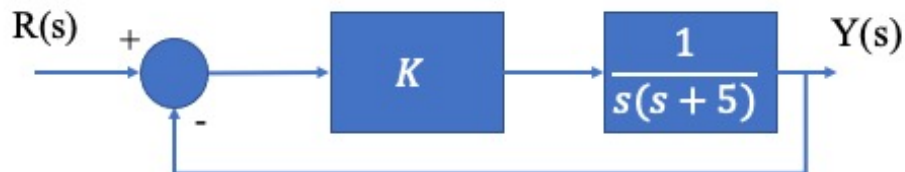


Figure 8: Unity feedback system for Q8.

(1 Point)

1. $0 \leq K \leq 20$
2. $5 \leq K \leq 25$
3. $K \geq 25$
4. $-1.5 \leq K \leq 10$

Answer:

10. Q10. A certain system is represented by the Bode diagram (only the magnitude in dB) shown in Fig. 9 (no zeros in the RHP). **Which is the correct answer showing the response of this system to a unit step input (assuming zero initial conditions)?**

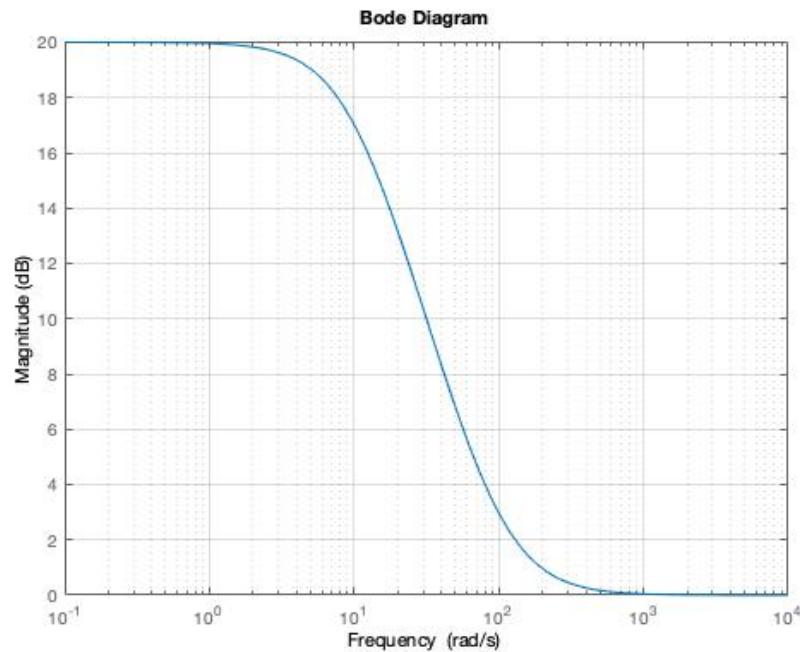
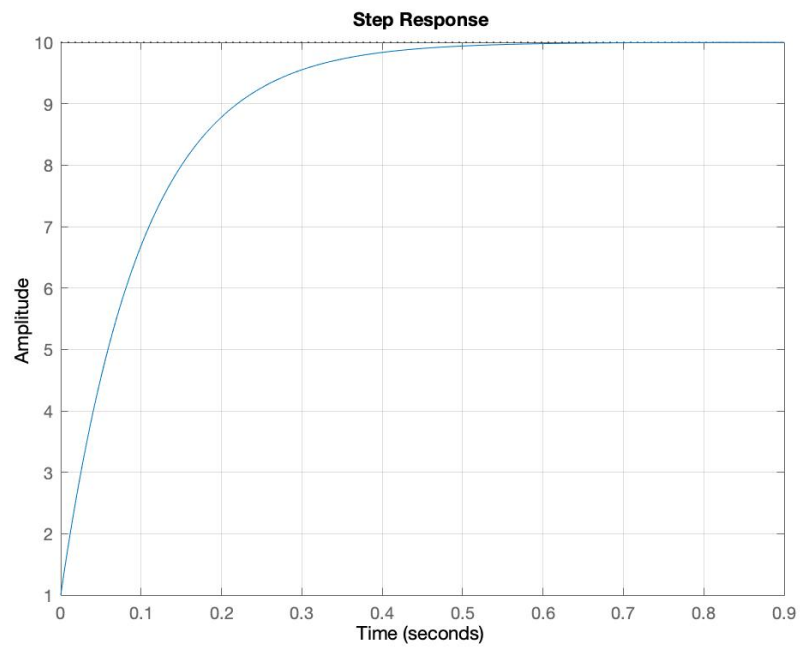
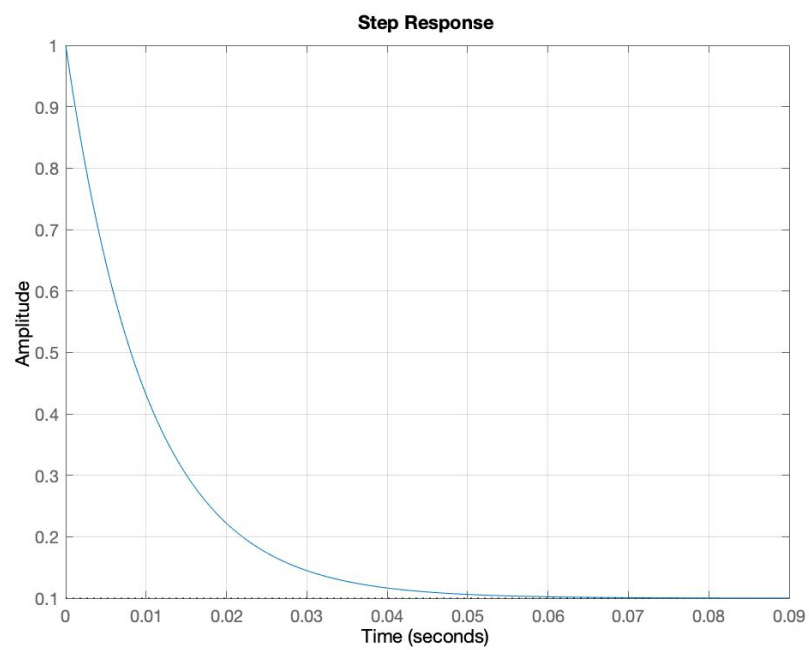


Figure 9: Magnitude portion (dB) of the Bode diagram for Q10.

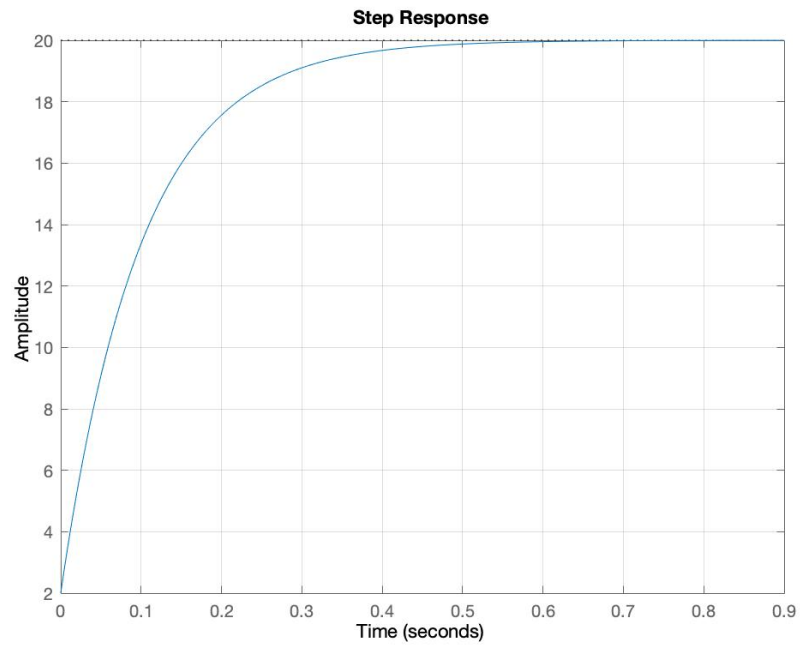
(1 Point)



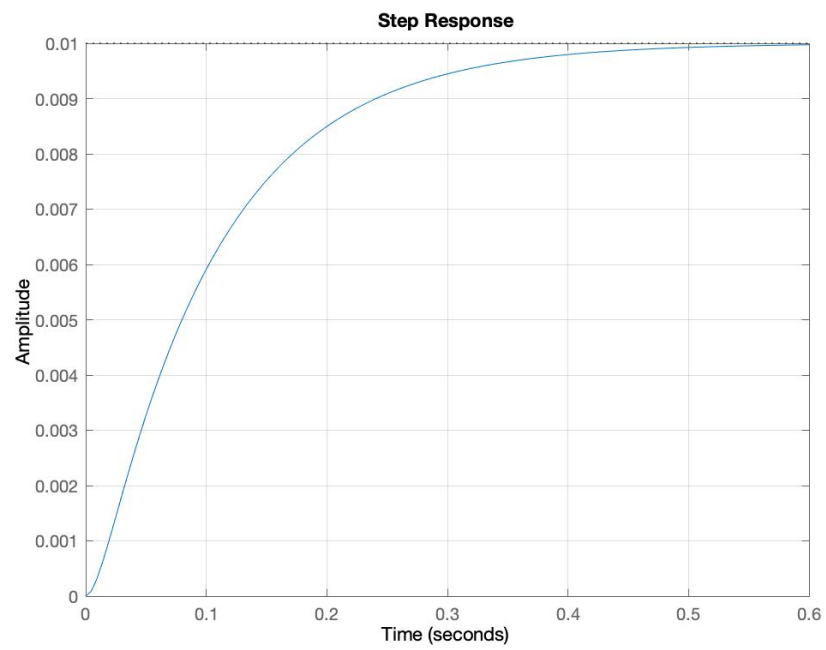
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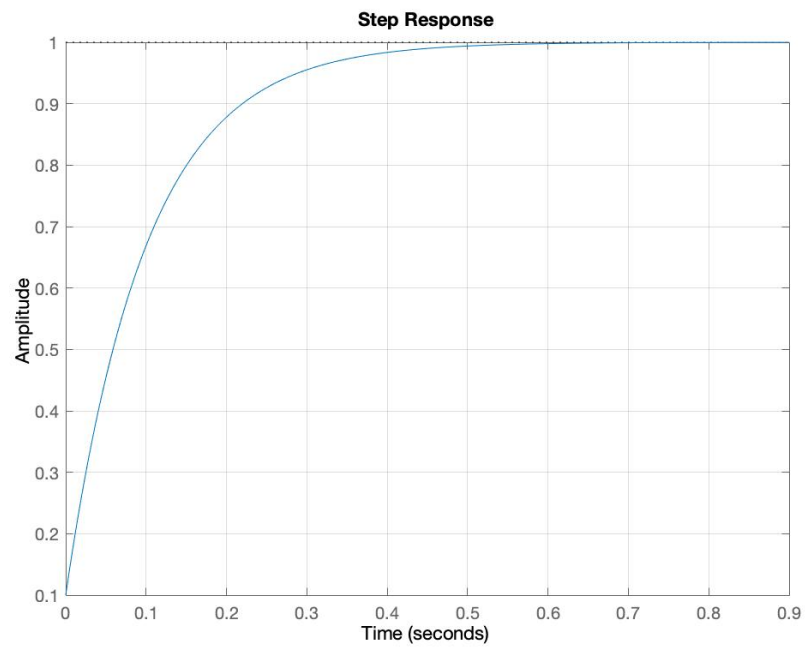
2.



3.



4.



5.

Answer: